

Facilities, Equipment, and Other Resources

NSF CISE CORE (IIS) - CopernicusAI Knowledge Engine

Principal Investigator: Gary Welz

Project Period: 36 months (3 years)

1. Facilities

Primary Work Location

Address: 460 West 24th St, Suite 3A, New York, NY 10011, United States

Facility Type: Home office / Remote work environment

Description: The Principal Investigator operates from a dedicated home office space suitable for software development, system design, research activities, and project management. The facility provides: - Secure workspace for development and research activities - High-speed internet connectivity for cloud-based development - Climate-controlled environment for equipment - Privacy for confidential project work and user study data

Access: 24/7 access for flexible development and research schedule

Adequacy: The facility is adequate to support all project activities including software development, system integration, user study coordination, and project management.

2. Equipment

Development Equipment

Primary Development System: - High-performance laptop computer (to be upgraded with Year 1 funds) - Current system supports development; upgrade will enhance capabilities for complex system integration - Used for: Code development, system design, testing, documentation, user study coordination

Peripheral Equipment: - External monitors for multi-screen development - Backup storage devices for data preservation - Network equipment for reliable connectivity

Equipment Ownership: All equipment is owned by the Principal Investigator

Equipment Maintenance: Maintained by Principal Investigator with manufacturer support as needed

Adequacy: Current equipment is adequate for project needs, with planned upgrades to enhance development capabilities.

3. Cloud Infrastructure Resources

Google Cloud Platform (GCP)

Services Utilized: - **Cloud Run:** Serverless container deployment for backend services - **Firestore:** NoSQL database for metadata and application data - **Cloud Storage:** Object storage for media files, transcripts,

structured data, and research datasets - **Cloud Functions:** Event-driven processing and automation - **Vertex AI:** Enterprise-scale LLM deployment and orchestration - **Cloud Tasks:** Background job queuing - **Secret Manager:** Secure API key and credential management

Current Usage: - Production CopernicusAI platform running on GCP - Genome Logic Modeling Project (GLMP) data (biological processes) stored in Cloud Storage - Research Paper Metadata Database infrastructure on GCP - Science Video Database backend infrastructure

Project Enhancements: - Expanded storage for unified metadata objects - Increased compute capacity for enhanced processing and evaluation - Additional Vertex AI usage for multi-model orchestration and reliability mechanisms - Enhanced analytics infrastructure for user studies

Cost Model: Pay-as-you-go with budget allocation in project budget (\$78,000 over 3 years)

Adequacy: Existing GCP infrastructure provides scalable, production-ready infrastructure for all project activities.

Vercel (Frontend Hosting)

Services Utilized: - Next.js application hosting - CDN for static assets - Serverless function hosting

Current Usage: - Production CopernicusAI frontend hosted on Vercel - Science Video Database frontend

Cost Model: Subscription-based with budget allocation in project budget (\$3,000 over 3 years)

Adequacy: Vercel provides reliable, scalable frontend hosting for all project components.

4. Software and Tools

Development Tools

- **Version Control:** Git, GitHub
- **IDEs:** VS Code, development IDEs
- **Programming Languages:** Python, TypeScript, JavaScript
- **Frameworks:** FastAPI, Next.js, React
- **Database Tools:** Firestore management tools
- **Testing Tools:** Unit testing frameworks, integration testing tools

Third-Party AI/ML Services

- **Google Gemini 3** (via Vertex AI)
- **OpenAI GPT-4/GPT-3.5** (via API)
- **Anthropic Claude 3** (via OpenRouter)
- **ElevenLabs TTS** (text-to-speech for audio briefings)

Cost Model: API-based usage with budget allocation (\$16,000 over 3 years)

Research Data Sources

- **PubMed/NCBI API** access
- **arXiv API** access
- **NASA ADS API** access

- **Zenodo API** access
- **CORE API** access
- **Google Scholar** access
- **YouTube Data API** (for science video database)

Adequacy: Access to comprehensive research data sources enables all project objectives.

Monitoring and Analytics

- Cloud monitoring tools (GCP native)
 - Application performance monitoring
 - Usage analytics tools for user studies
 - Evaluation metrics collection systems
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5. Data and Information Resources

Existing Data Assets

- **GLMP Database:** Biological processes in Google Cloud Storage
- **Research Paper Metadata:** Prototype database structure
- **Science Video Database:** Transcript and metadata infrastructure
- **CopernicusAI Platform:** Generated podcast episodes with metadata

Access to Research Databases

- Integration with multiple academic databases
- Access to research papers via APIs
- Real-time research discovery capabilities

Adequacy: Existing data assets and database integrations provide strong foundation for project research.

6. Human Resources

Principal Investigator

- **Gary Welz:** 6 months/year effort (50% FTE equivalent)
- **Qualifications:** Retired faculty (Mathematics and Computer Science, John Jay College CUNY), extensive experience in software development, AI/ML systems, and research tool development
- **Role:** Research leadership, technical development, system design, project management, user study coordination

Support Personnel

- **Research Assistant/Software Developer:** 4-5 months/year (budgeted in project)
- **Student User Testers:** Graduate and undergraduate students (budgeted: \$80,160 over 3 years)
- **Technical Consultants:** Specialized expertise as needed (budgeted: \$17,000 over 3 years)

Adequacy: PI has necessary qualifications and experience, with adequate support resources budgeted.

7. Academic Affiliations

CUNY Graduate Center - New Media Lab

- **Status:** Affiliate member (retired faculty status)
- **Resources:** Access to academic resources, collaboration opportunities, research community
- **Note:** Academic affiliation provides research context but project is independent research

John Jay College of Criminal Justice

- **Status:** Retired faculty, Mathematics and Computer Science Department
- **Background:** Provides research and teaching experience context

Adequacy: Academic affiliations provide research context and potential collaboration opportunities.

8. Resource Management

Cloud Resource Management

- **Monitoring:** Continuous monitoring of cloud resource usage
- **Optimization:** Regular review and optimization of cloud costs
- **Scaling:** Ability to scale resources up/down based on project needs
- **Budget Controls:** Budget alerts and limits to prevent cost overruns

Equipment Management

- **Maintenance:** Regular maintenance of development equipment
- **Upgrades:** Planned equipment upgrades in Year 1 (budgeted: \$8,000)
- **Security:** Physical and digital security for all equipment

Data Management

- **Backup:** Automated backup systems for all critical data
 - **Version Control:** Git-based version control for all code
 - **Documentation:** Comprehensive documentation of all systems and processes
 - **Data Preservation:** Long-term preservation strategy (project duration + 3 years)
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9. Resource Adequacy Statement

The facilities, equipment, and resources described above are adequate to support all project objectives:

[OK] **Development Environment:** Home office provides suitable workspace for software development and research activities

[OK] **Cloud Infrastructure:** Existing GCP and Vercel infrastructure supports production-scale development and research

[OK] **Equipment:** Current equipment adequate, with planned upgrades to enhance capabilities

[OK] **Software Tools:** Access to all necessary development tools, frameworks, and AI/ML services

[OK] **Data Resources:** Existing integrations with research databases provide necessary data access

[OK] **Human Resources:** PI has necessary qualifications and experience, with adequate support resources budgeted

[OK] **Scalability:** Cloud-based infrastructure allows for scaling as project needs grow

[OK] **User Studies:** Infrastructure and resources adequate to support alpha/beta testing programs with student participants

10. Resource Sharing and Collaboration

Open Source Components

- Code and tools will be shared via GitHub (MIT License)
- Documentation will be publicly accessible
- Metadata schemas will be publicly available
- Non-proprietary components available to research community

Proprietary Components

- Business-sensitive algorithms and implementations remain proprietary
 - Commercial rights retained for proprietary components
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Document Prepared By: Gary Welz, Principal Investigator

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